

Fault-tolerant Bayesian Decentralized Data Fusion Using Reliability Variables and Mixture Models*

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Abstract—In uncertain and dynamic environments, decentralized data fusion (DDF) techniques have been widely used to estimate the states and the uncertainty levels over large mission spaces in a robust and scalable way. In data fusion frameworks using distributed sensor networks, undetected sensor failures can degrade the quality of fusion results of the entire system. Therefore, DDF methods which are robust to inconsistent data are needed. In this paper, a fault-tolerant Bayesian DDF method using Gaussian mixture models is developed. The probability of agent reliability states, which represent consistency of local estimates that agents share with their neighbors, are modeled as weights of mixture models and estimated together with the target process. The target process and reliability states are updated in a decentralized Bayesian way, exploiting the properties of Gaussian mixture models. To prevent the hypothesis explosion problem of Gaussian mixture models, a Runnalls' method with consideration of physical meaning of mixture weights is utilized. A numerical simulation on a 2D dynamic target tracking problem is presented to verify performance of the suggested algorithm and compared with existing DDF methods. It is shown that the suggested algorithm gives robust and less conservative fusion results than the others.

Index Terms—State estimation, Distributed inference, Multi-robot systems, Distributed sensor network, Fault-tolerant Bayesian DDF

I. INTRODUCTION

In uncertain and dynamic environments, it is necessary for autonomous vehicles to precisely estimate their own states and the states of their environments, as well as quantify uncertainty to ensure best overall performance. For estimation tasks, fusing information from multiple distributed sensors in a network enables effective and scalable information gathering. A centralized data fusion strategy enables ideal fusion results, but it suffers from the inherent issues of single point of communications and computation failures. To overcome these issues and make the entire system more scalable, decentralized data fusion (DDF) methods have been suggested. In this sense, a wide range of data fusion algorithms, including Bayesian DDF [1,2], Dempster-Shafer inference [3,4], and fuzzy logic

[5,6] have been widely studied. Among these variety of algorithms, Bayesian DDF is capable of tracking the full state distribution of a given process, and can easily and recursively fuse new data shared across multiple platforms with prior knowledge, without the need to store or communicate massive amounts of raw sensor data [2]. These properties and its rigorous theoretical framework have made Bayesian DDF a well-established technique for decentralized state estimation.

Due to aforementioned advantages, multi-robot system architectures are becoming increasingly popular to information gathering tasks. However, major concerns can arise if some robots or their sensors fail during the mission. Once a sensor on a certain platform fails, inconsistent data or estimates are generated, and communicated to the others. If other fusing agents do not know about and account for this inconsistency, the quality of overall fusion result can be quite degraded. As such, the sensor failure problem can affect the credibility of fusion results. However, despite the potential failure, there has been lack of research dealing with fault-tolerant Bayesian fusion methods. Therefore, the goal of this study is to develop an effective Bayesian DDF algorithm, that is robust to inconsistent data.

One strategy for adapting Bayesian data fusion is to extend conservative fusion strategies like Covariance Intersection (CI) [7]. Since CI guarantees conservative fusion result only if the individual estimates are both consistent, methods like Covariance Union (CU) [8] and Generalized Covariance Union (GCU) [9] try to account for all possible multiple failure hypotheses among the networked platforms, so that agents in the network “hedge” their estimates against potentially inconsistent estimates received by others. Since CU and GCU cover both of potentially inconsistent estimates, these methods naturally give overly conservative fusion results compared to CI and exact Bayesian methods. The authors in [8] suggested utilizing CI after CU to reduce this large uncertainty, but this approach cannot deal with persistent error case (i.e., once a

sensor on the platform fails, it stays in failed state afterward). More explicit strategies that model failure probabilities deploy methods akin to those used for data association, by leveraging mixture models to quantify the uncertainty associated with different agents being in different failure/non-failure states over time. This idea can be combined with Standard Mixture Reduction (SMR) [10] techniques to give less conservative fusion results compared to CU and GCU, but this method requires prior knowledge of each agent's reliability.

The other way of dealing with inconsistent data in fusion framework is to detect erroneous measurement and eliminate it from fusion process. To find the faulty agent, error detection module (EDM) compares outputs of local fusion blocks and utilize similarity between them as determination metric [11]. Also, a method to eliminate inconsistent data by analyzing the variance level of fusion results, which change according to the consistency of local measurement, has been suggested [12]. However, these deterministic methods using EDM and elimination process are highly dependent on setting of the elimination criteria, and hard to apply for decentralized case, since they inherently require comparisons between local data.

In this study, a fault-tolerant Bayesian DDF method which considers online estimation of agent reliability states is developed. The agent reliability here means that consistency of local estimate that agent shares to its neighbors. The joint distribution over the target process and the reliability states is modeled by Gaussian mixture models, where the probability of different reliability state hypotheses serve as mixture weights. The augmented state vector (defined by the target process and the reliability states) is conditionally factorized, so that the target process and the reliability states can be separately updated. In this process, two distinct channel filters (CF) [1] are utilized to exactly track common information, which enables decentralization of the fusion procedure. The goal of this algorithm is to have robustness to data inconsistency, and give less conservative fusion results compared to existing fault-tolerant fusion methods, with the capability to deal with persistent error cases.

II. PROBLEM STATEMENT

The decentralized data fusion problem considered in this paper is defined as follows, described by simple two fusing agent case for the ease of explanation. Agent i and j estimate a common target process, x . Along with the target state, each agent has its own reliability state r^i and r^j , so the augmented state vector at time k can be defined as

$$s_k = [x_k, r_k] = [x_k, r_k^i, r_k^j]. \quad (1)$$

The joint reliability state r is represented by combinations of binary local reliability states r^i and r^j , and r can have value among 11, 10, 01, and 00.

The agent reliability states may change at every time step, and affect the local measurement on the target process at time k , z_k^i and z_k^j . Each agent calculates local posterior distribution on s using its prior and local measurement, and shares this

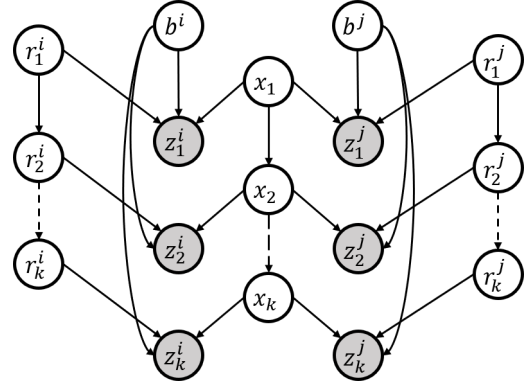


Fig. 1. Bayesian network. White nodes are unobserved and shaded nodes are observed random variables.

posterior with its neighboring agents. Therefore, if we define cumulative information set at time k from agent i as Z_k^i , the goal of the decentralized data fusion is to find the fusion rule:

$$p_f(s) = p(s|Z_k^i \cup Z_k^j) = \mathbb{F}(p(s|Z_k^i), p(s|Z_k^j)) \quad (2)$$

It is assumed throughout this paper that a “fault” in one agent results in its local sensor measurement on target process becoming corrupted, so that its local state estimates are no longer reliable. If θ^i denotes failure probability of agent i at each time step, the reliability state of agent i at time k is modeled as:

$$\begin{cases} r_k^i = 0 & \text{if } r_{k-1}^i = 0 \\ r_k^i = \text{Bernoulli}(1 - \theta^i) & \text{if } r_{k-1}^i = 1 \end{cases} \quad (3)$$

Equation (3) means reliable agent i can become unreliable ($r^i = 0$) with probability θ^i , and once a sensor on the platform fails during the operation, it cannot be recovered and collects inaccurate data from that point on. This error model is denoted as “persistent error” hereafter. Depending on its reliability state, an agent makes different measurements on the target process. If an agent is unreliable, a known bias b is added to an otherwise accurate measurement. Then unknown true target and reliability states are estimated by using acquired measurements.

Fig. 1 represents Bayesian network for dynamic target process. Since the reliability states of different agents do not affect each other, it can be assumed that reliability states of fusing agents are mutually independent ($r_k^i \perp\!\!\!\perp r_k^j$). Also, the local measurement of each agent is determined by the true target state, its true reliability state and bias. Therefore, it can be naturally assumed that measurements of fusing agents are conditionally independent given target state ($z_k^i \perp\!\!\!\perp z_k^j | x_k$), which is reasonable for general distributed information gathering problems.

III. FAULT-TOLERANT BAYESIAN DDF

A. Fault-tolerant Fusion using Gaussian Mixtures

In this paper, fault-tolerant Bayesian DDF method using Gaussian mixture models is suggested. The online reliability

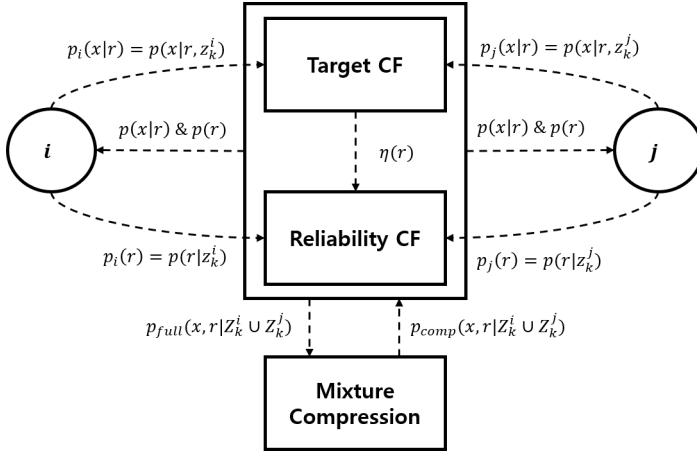


Fig. 2. Overall fusion procedure

states of fusing agents are defined as variables to be estimated. The joint distribution on the augmented state vector s is conditionally factorized, and modeled by Gaussian mixture models. The reliability state probabilities are utilized as weights of Gaussian mixture models, representing likelihood of various reliability hypotheses.

By modeling the joint distribution on the target process and the reliability states through Gaussian mixture models, the estimation procedure can be separated into estimation of continuous target process and of discrete reliability states. This makes the estimation problem simpler, and also well-established Bayesian filters for continuous variables can be used for target process estimation. To decentralize the estimation process on the target and the reliability states, two distinct CFs [1] are utilized to track the common information exactly. Since the joint distribution is represented by multiple Gaussian mixands, the message from target state CF to reliability state CF (denoted by $\eta(r)$ in Fig. 2) can be exactly calculated, so that the decentralized fusion process becomes tractable. A mixture compression method based on well-known Runnalls' method [13] is utilized to prevent "hypothesis explosion" problem, with some modification to deal with mixture weights physically representing the reliability states of agents, and have their own dynamics.

The overall fusion procedure can be represented as Fig. 2. Each local agent gets common information on target process and reliability states from two CFs, and updates common information by using its own local measurement. Then locally updated posterior distributions are fused in a decentralized way at distinct CFs, along with exact calculation of $\eta(r)$. Finally, fused joint posterior distribution on target process and reliability states is compressed, and the compressed mixture models are used as common information at the next time step.

B. Conditional Factorization

When $s = [x, r]$, we can conditionally factorize joint distribution over s as

$$p(s) = p(x, r) = p(x|r)p(r) \quad (4)$$

Since decentralized fusion rule can be written as

$$p_f(s) = p(s|Z_k^i \cup Z_k^j) \propto \frac{p(s|Z_k^i)p(s|Z_k^j)}{p(s|Z_k^i \cap Z_k^j)} = \frac{p_i(s)p_j(s)}{p_c(s)} \quad (5)$$

where $p_i(\cdot)$ denotes local posterior distribution from agent i , and $p_c(\cdot)$ denotes common prior distribution of agent i and j .

By applying (4) to (5), we can conditionally factorize joint posterior distribution over s as shown in [14],

$$p_f(s) \propto \frac{p_i(x|r)p_i(r)p_j(x|r)p_j(r)}{p_c(x|r)p_c(r)} = \frac{p_i(x|r)p_j(x|r)}{p_c(x|r)} \cdot \frac{p_i(r)p_j(r)}{p_c(r)}. \quad (6)$$

Let $\tilde{p}_f(x|r)$ be unnormalized fused conditional pdf,

$$\tilde{p}_f(x|r) = \frac{p_i(x|r)p_j(x|r)}{p_c(x|r)}. \quad (7)$$

If we normalize this and define $\eta(r)$ as

$$p_f(x|r) = \tilde{p}_f(x|r) \cdot \frac{1}{\eta(r)} \quad \eta(r) = \int \tilde{p}_f(x|r) dx, \quad (8)$$

then we can rewrite (6) as

$$p_f(s) \propto p_f(x|r)\eta(r)p_f(r). \quad (9)$$

Now we separated Bayesian update process of augmented state vector s into separate updates of the conditional target distribution and the reliability state distribution. We can update the conditional target distribution in a decentralized way using parallel CFs for each reliability hypotheses. Also if we can calculate $\eta(r)$, reliability posteriors can be fused in a decentralized way by using another CF.

C. Local Posterior Update

Suppose we have four different conditional prior distributions, $p(x|r_{k-1})$ for joint reliability hypotheses (11, 10, 01, 00) of fusing agents at previous time step. From the perspective of agent i , we can calculate local posterior distribution over x corresponding to previous joint reliability hypothesis r_{k-1} and current joint reliability hypothesis r_k by using Bayes' rule,

$$p_i(x|r_{k-1}, r_k) = p(x|r_{k-1}, r_k, z_k^i) = \frac{p(z_k^i|x, r_{k-1}, r_k)p(x|r_{k-1}, r_k)}{p(z_k^i|r_{k-1}, r_k)} = \frac{p(z_k^i|x, r_{k-1}, r_k^i)p(x|r_{k-1}, r_k)}{p(z_k^i|r_{k-1}, r_k^i)}. \quad (10)$$

As written in the last line of (10), agent i can update its local posterior only considering its own current reliability hypothesis r_k^i , since agent i 's local measurement is independent to agent j 's reliability state ($z_k^i \perp r_k^j$).

Likewise, the local posterior on joint reliability states can be calculated again using Bayes' rule,

$$\begin{aligned} p_i(r_{k-1}, r_k) &= p(r_{k-1}, r_k | z_k^i) \\ &= \frac{p(z_k^i | r_{k-1}, r_k) p(r_{k-1}, r_k)}{p(z_k^i)} \\ &= \frac{p(z_k^i | r_{k-1}, r_k^i) p(r_{k-1}, r_k)}{p(z_k^i)}. \end{aligned} \quad (11)$$

Although it is not specified in (10) and (11), prediction steps of the target and the reliability states are conducted before measurement updates at each local estimators. The prediction and measurement update of the target process, corresponding to (10), can be conducted together by using Bayesian filters in a parallel manner, according to the reliability hypotheses. Also, normalization constant $p(z_k^i | r_{k-1}, r_k^i)$ from (10), which will be obtained during local target posterior calculation, can be reused as measurement likelihood while updating local reliability posterior.

D. Decentralized Posterior Fusion

Calculated local posterior distributions over target process and joint reliability state can be fused at distinct CFs, target CF and reliability CF. As can be seen from (9), $\eta(r)$ is required to calculate fused joint distribution of augmented state. Unfortunately, calculation of $\eta(r)$ for general pdfs is not an easy task, since it includes integration over all possible x 's. For the case of fusing Gaussian pdfs of static process, or dynamic process when local estimators and target CF have same system model, it is possible to analytically calculate $\eta(r)$ at target CF.

The Gaussian pdf in information form is,

$$\mathcal{N}^{-1}(x; \nu, \Lambda) \triangleq \frac{\exp(-\frac{1}{2}\nu^\top \Lambda^{-1} \nu)}{(2\pi)^{n/2} \sqrt{\det \Lambda^{-1}}} \exp\left(-\frac{1}{2}x^\top \Lambda x + x^\top \nu\right). \quad (12)$$

The Channel filter [1] is a method to exactly track common information, $p_c(\cdot)$ between two fusing agents. The CF recursively updates the common information between two fusing agents, by propagating and comparing the predicted distribution with the local posteriors. The Gaussian CF update equation in information form can be written as,

$$\begin{aligned} \nu_f &= \nu_i + \nu_j - \nu_c \\ \Lambda_f &= \Lambda_i + \Lambda_j - \Lambda_c. \end{aligned} \quad (13)$$

where subscript i and j mean local posterior from each agent, c means (predicted) common prior in target CF, and f means fused posterior.

From (7) and (8), $\eta(r)$ can be rewritten as,

$$\begin{aligned} \eta(r) &= \frac{\tilde{p}_f(x|r)}{p_f(x|r)} \\ &= \frac{p_i(x|r)p_j(x|r)}{p_c(x|r)} \cdot \frac{1}{p_f(x|r)}. \end{aligned} \quad (14)$$

When both of local posteriors, common prior, and fused posteriors are all Gaussian distributions, $\eta(r)$ can be exactly calculated by using (12). By canceling out common terms, $\eta(r)$ can be simply represented as,

$$\begin{aligned} \eta(r) &= \sqrt{\frac{\det \Lambda_i \det \Lambda_j}{\det \Lambda_c \det \Lambda_f}} \\ &\exp \left\{ -\frac{1}{2} \left(\nu_i^\top \Lambda_i^{-1} \nu_i + \nu_j^\top \Lambda_j^{-1} \nu_j - \nu_c^\top \Lambda_c^{-1} \nu_c - \nu_f^\top \Lambda_f^{-1} \nu_f \right) \right\}. \end{aligned} \quad (15)$$

Now, by using (10) - (15), local posterior distributions of x and r can be calculated, and fused in a decentralized way.

E. Mixture Compression

From a single prior distribution on the reliability hypothesis, there can be four posterior distributions of current reliability hypotheses, similar to Gaussian sum filters with non-Gaussian measurement noise. Therefore, if we form a Gaussian mixture by marginalizing (4) after fusion with respect to r , the number of mixands increases geometrically every time step, which is called "hypothesis explosion" problem. To resolve this issue, a mixture compression method based on Runnalls' method [13] is utilized.

Runnalls' method is a mixture compression method that iteratively merges two "closest" Gaussian distributions to make the number of mixands as desired. At each merge round, the upper bound of Kullback-Leibler divergence (KLD) between pre-merge and post-merge Gaussian mixtures, $D(\alpha, \beta)$ is calculated for all possible (α, β) pairs. $D(\alpha, \beta)$ is calculated as,

$$\begin{aligned} D(\alpha, \beta) &= \frac{1}{2} [(w_\alpha + w_\beta) \log \det(P_{\alpha\beta} - w_\alpha \log \det(P_\alpha) \\ &\quad - w_\beta \log \det(P_\beta))] \\ P_{\alpha\beta} &= \frac{w_\alpha}{w_\alpha + w_\beta} P_\alpha + \frac{w_\beta}{w_\alpha + w_\beta} P_\beta \\ &\quad + \frac{w_\alpha w_\beta}{(w_\alpha + w_\beta)^2} (m_\alpha - m_\beta)(m_\alpha - m_\beta)^\top, \end{aligned} \quad (16)$$

where m , P , and w denote mean, covariance matrix, and weight of mixand respectively.

Unlike other general cases using Gaussian mixture models, the mixture weight in our problem has physical meaning of the likelihood of each reliability hypothesis. Therefore, the prediction step of the mixture weights should reflect dynamics of the agent reliability states. The true dynamics of agent reliability state is defined as (3). Therefore, by utilizing prior knowledge of agent failure mode ("persistent") and failure

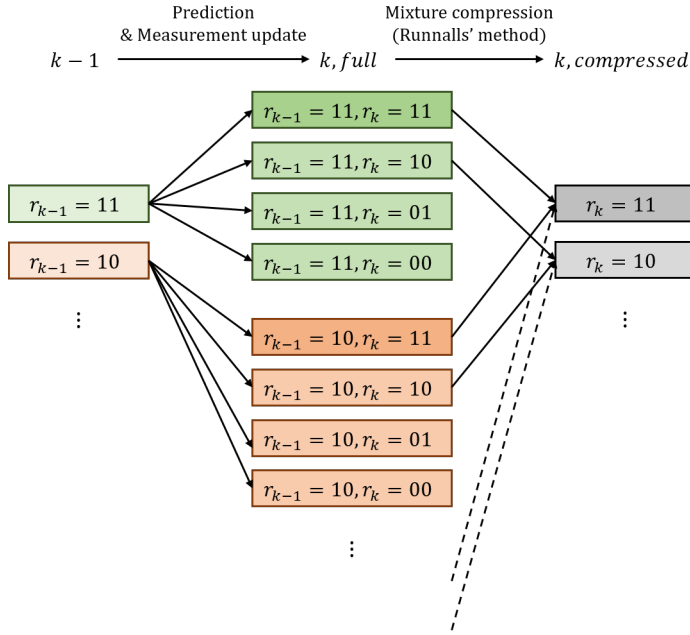


Fig. 3. Mixture compression procedure

probability θ of each agent, the prediction step of reliability states can be represented as,

$$p(r_k|z_{k-1}) = F_r p(r_{k-1}|z_{k-1})$$

$$F_r = \begin{bmatrix} (1-\theta^i)(1-\theta^j) & 0 & 0 & 0 \\ (1-\theta^i)\theta^j & 1-\theta^i & 0 & 0 \\ \theta^i(1-\theta^j) & 0 & 1-\theta^j & 0 \\ \theta^i\theta^j & \theta^i & \theta^j & 1 \end{bmatrix}. \quad (17)$$

To apply (17), compressed mixture weights should retain their physical meaning, the likelihood of current reliability hypotheses. Therefore, Runnalls' method is applied to mixands of same "current" hypothesis (i.e. merging four mixands of different "previous" hypotheses) in a parallel manner for each current hypothesis. It can be interpreted as filtering previous trajectories of reliability states. The overall mixture compression procedure can be represented as fig. 3.

The overall procedure of fault-tolerant Bayesian DDF can be represented as Algorithm 1.

IV. NUMERICAL SIMULATION RESULT

A. Simulation Setup

For verification of the suggested algorithm, numerical simulation is conducted on a 2D dynamic target tracking problem with two agents (denoted by agent 1 and 2). The target process x and joint reliability state r are defined as,

$$x = [e, \dot{e}, n, \dot{n}]^\top$$

$$r = [r_1, r_2]^\top, \quad (18)$$

where e and n denotes position in easting/northing directions, and $\dot{e}$ and $\dot{n}$ denotes velocity.

Algorithm 1 Fault-tolerant Bayesian DDF

Define prior pdfs on x and r

for All time steps **do**

Propagate target process in local estimators

Make local measurements

Update local posterior on x and $r \triangleright$ Eq. (10) and (11)

Propagate target process in target CF

Fuse local posteriors on x in target CF

Calculate $\eta(r)$ and send to reliability CF $\triangleright$ Eq. (15)

Propagate reliability states in reliability CF $\triangleright$ Eq. (17)

Fuse local posteriors on r in reliability CF

Compress Gaussian mixtures

Joint prior, $p(x, r) \leftarrow$ Compressed Gaussian mixtures

end for

For the simplicity, the target has linear constant velocity dynamics, and position of the target is directly measured by sensors. However, the suggested algorithm can be applied to a nonlinear problem without loss of generality by replacing local estimators by nonlinear filters, like the unscented Kalman filter. The target dynamics can be written as,

$$x_{k+1} = F_t x_k + G_t u_k + w_k \quad (19)$$

where

$$u_k = [1, 0]^\top$$

$$w_k \sim \mathcal{N}(0, Q_t)$$

$$F_t = \begin{bmatrix} 1 & dt & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & dt \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad G_t = \begin{bmatrix} \frac{1}{2}dt^2 & 0 \\ dt & 0 \\ 0 & \frac{1}{2}dt^2 \\ 0 & dt \end{bmatrix}$$

$$Q_t = \begin{bmatrix} 0.01 & 0 & 0 & 0 \\ 0 & 0.04 & 0 & 0 \\ 0 & 0 & 0.01 & 0 \\ 0 & 0 & 0 & 0.04 \end{bmatrix}.$$

The measurement model of agent i can be described as,

$$z_{k+1}^i = H^i x_{k+1} + (1 - r_{k+1}^i) b^i + v_{k+1}^i \quad (20)$$

where

$$v_{k+1}^i \sim \mathcal{N}(0, R^i)$$

$$H^1 = H^2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$R^1 = R^2 = \begin{bmatrix} 0.36 & 0 \\ 0 & 0.36 \end{bmatrix}.$$

Since the goal of the suggested algorithm is to give consistent and compact fusion result in spite of inconsistent data, it is assumed that at least one fusing agent makes consistent local estimate. Therefore, for the clarity of analysis, agent 1 is set to stay normal, while agent 2 follows failure model defined by (3) with the failure probability $\theta^2 = 0.05$. As can be seen

from (20), an additive bias b is added to otherwise accurate but noisy measurement when the true reliability state of an agent is 0. In this case, b is set to be $[2, -2]^\top$.

The same prior distribution on the target process is used for all reliability hypotheses. The target and reliability priors are set to be,

$$\begin{aligned} p_{pr}(x|r) &= \mathcal{N}(\mu_{pr}, P_{pr}) \\ p_{pr}(r) &= [(1-\theta)^2, \theta(1-\theta), \theta(1-\theta), 0]^\top \\ \mu_{pr} &= x_0 \\ P_{pr} &= 4I_4 \\ \theta &= 0.05, \end{aligned}$$

where x_0 means initial target state vector.

To update two CFs for target process and reliability states, it is assumed that both of local estimators and target CF have same dynamics model, which guarantee the proper tracking of the common information for dynamic process.

To verify fusion performance of the suggested algorithm, CU/CI [15] and Bayesian DDF method without considering agent reliability (denoted as “naive Bayesian”) were used for comparison. Both of algorithms used same local estimators as the suggested one. For all cases, it is assumed that local agents have no information of the control input of the target.

B. Result and Discussion

Fig. 4 represents the fusion result of the suggested algorithm and comparison group. The result of the suggested algorithm, denoted by FT Bayesian, is represented by blue line. The red line represents CU/CI and the yellow line represents naive Bayesian DDF method.

Fig. 4(a) shows the true position trajectory and the position estimates of all algorithms. It is shown that the suggested algorithm can track the target state properly, despite the existence of agent failure at time $k = 12$ (from the true reliability history plot of Fig. 4(d)). Due to the inconsistent local estimate from faulty agent 2, it is found that the fusion result of naive Bayesian DDF is shifted to the direction of the measurement bias of agent 2. Since CU/CI makes fusion result to cover both of local estimates (which are potentially inconsistent), the fused estimate of CU/CI also shifted to the same direction. This phenomenon can also be seen in the absolute error plot of position estimate in Fig. 4(b). Similar to Fig. 4(a), the suggested algorithm showed the smallest error overall, and CU/CI and naive Bayesian DDF showed very similar behaviors in terms of point estimate.

Fig. 4(c) shows the trace of error covariance matrix, which represents the uncertainty level of the fusion result. All algorithms showed similar value before the agent 2 fails, but CU/CI showed much larger level of uncertainty after the failure compared to the others. The naive Bayesian DDF showed similar value with the suggested algorithm, but there was shift in the point estimate which means that, fusing without considering the reliability states can lead to inconsistency (and this is the reason that CU/CI gives fused estimate with larger

uncertainty). Fig. 5 represents the NEES chi-square consistency test result of the suggested algorithm with $\alpha = 0.05$. The NEES statistics of the fusion result fall into the boundary for more than 95% of the total estimation time. As can be seen from Fig. 4(c) and Fig. 5, our Bayesian DDF algorithm can give consistent and compact fusion result compared to existing methods, in spite of inconsistent local estimate.

Fig. 4(d) shows the history of true and estimated reliability states. By using the suggested algorithm, the reliability states can be properly estimated in general. However, there is instability of reliability estimates especially in the early stage. This instability can make the fusion result to be inaccurate, so improving this reliability estimation process should be a future work.

A summary of the results is as follows. The fusion performance of the suggested fault-tolerant Bayesian DDF algorithm is compared with existing methods for the problem where agent reliability changes during the system operation. In terms of the accuracy of the point estimate, the suggested algorithm showed better performance than comparison group. Also, the suggested algorithm can give less conservative fusion result compared to CU/CI, while maintaining its consistency. It was also able to properly estimate reliability states, which is one of the key parts to track the target in the presence of inconsistent local estimate.

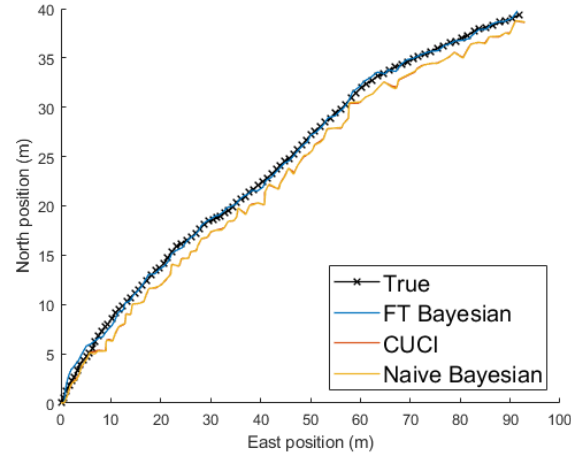
V. CONCLUSION

In this paper, a fault-tolerant Bayesian DDF algorithm using agent reliability variables and Gaussian mixture models was developed. The joint reliability states of fusing agents is used as mixture weights, and locally estimated together with the target process. The local estimates are fused in a decentralized manner, and this process could become tractable by using the advantages of Gaussian mixture models. To prevent hypothesis explosion problem, Runnalls’ method considering the physical meaning and dynamics of weights was utilized to compress the Gaussian mixtures.

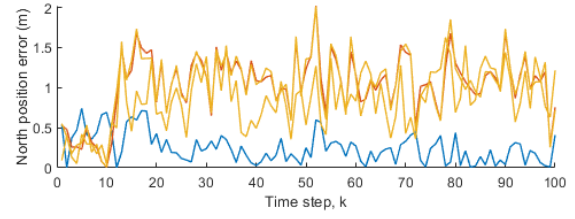
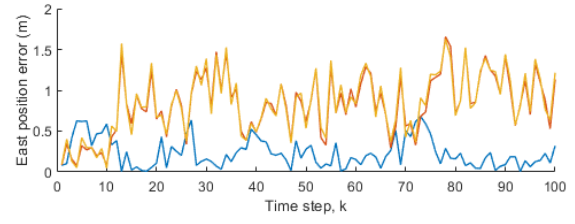
To verify fusion performance of the suggested algorithm and compare with the existing methods, numerical simulation on a 2D dynamic target tracking problem was conducted. It is shown that the suggested algorithm can appropriately estimate both of target process and reliability states, despite sensor failure during the operation. While CU/CI and Bayesian DDF without considering agent reliability led to biased fusion results due to inconsistent local estimate, our algorithm could track the target more accurately. Also in terms of uncertainty, less conservative but still consistent fusion result can be made compared to CU/CI, and the suggested algorithm could deal with persistent errors properly.

The future research directions from this study are as follows:

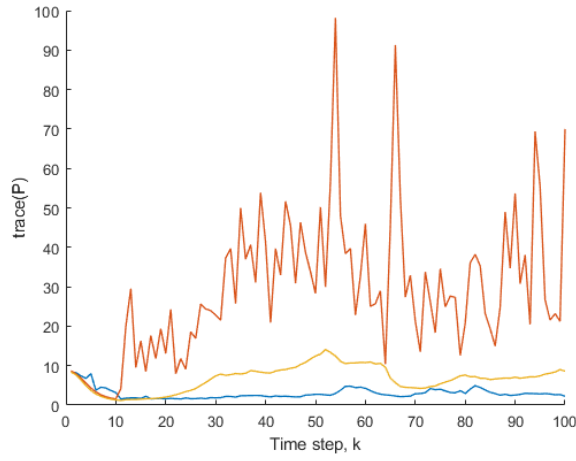
- Generalize for unknown/partially known erroneous measurement model (e.g. the amount of measurement error increases since the first failure occurred): need to determine how accurate prior knowledge for erroneous measurement model is required, and improvement and



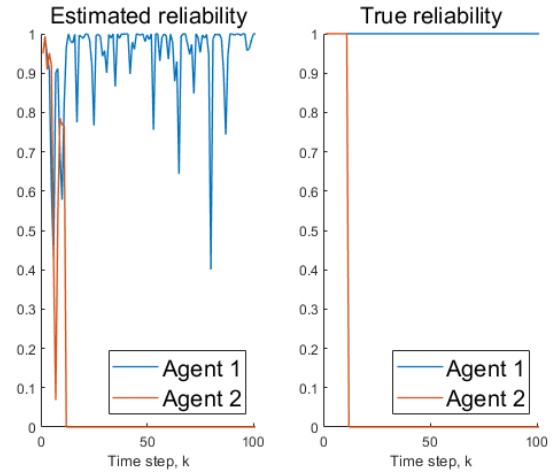
(a)



(b)



(c)



(d)

Fig. 4. (a) Position estimates (b) Absolute position error (c) Trace of error covariance matrix (d) Reliability estimates

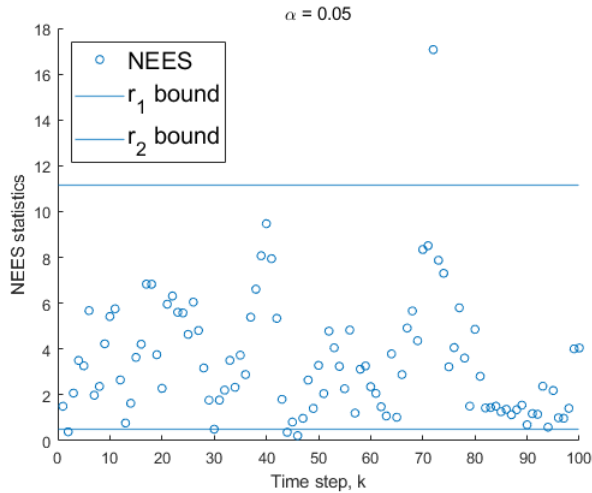


Fig. 5. NEES chi-square test result $\alpha = 0.05$

stabilization of reliability estimate method is needed for unknown measurement error cases.

- Heterogeneous data fusion: factorize state variables according to conditional independence to reduce communication overhead [16].
- More realistic simulation and real-world demonstration: realistic simulation considering system nonlinearities in dynamics and measurement model and complex network topology, and real-world demonstration on robotic platforms.

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